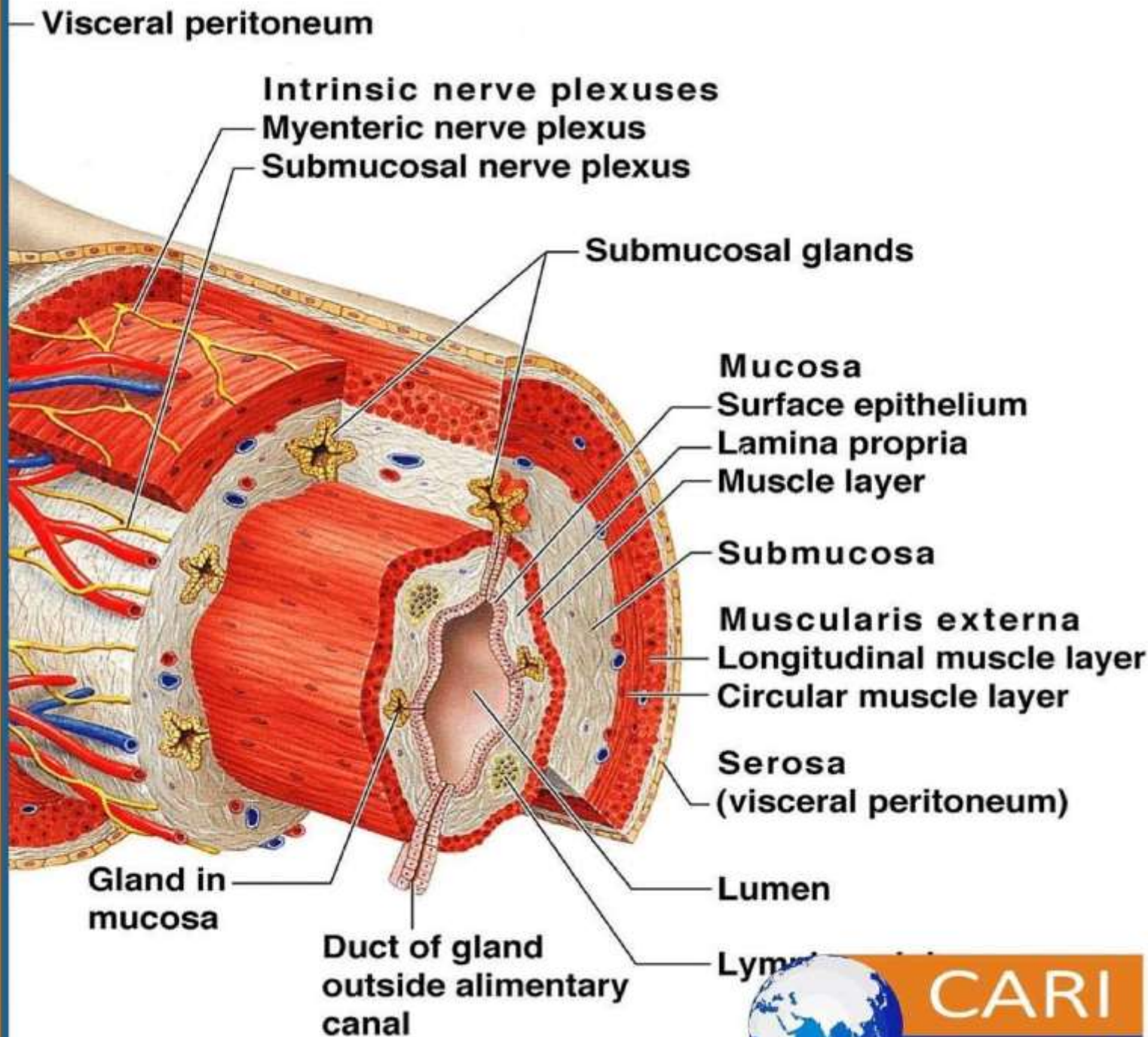


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A Nano-encapsulated Seven-strain Microbial Biostimulant improves Nutrient-use Efficiency, Soil Health and Crop Productivity across Contrasting Soil Orders: An Integrated Laboratory, Greenhouse and Multi-site Field Validation



A Nano-encapsulated Seven-strain Microbial Biostimulant improves Nutrient-use Efficiency, Soil Health and Crop Productivity across Contrasting Soil Orders: An Integrated Laboratory, Greenhouse and Multi-site Field Validation

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Abstract

Purpose: This study evaluated the agronomic performance, stability, soil-health effects and climate-resilience potential of a nano-encapsulated microbial biostimulant designed to improve fertilizer-use efficiency and reduce nutrient losses. The formulation combined a seven-strain microbial consortium with a chitosan nanocarrier co-loaded with molybdenum, silicon and plant-growth regulators.

Methodology: A fully integrated 12-month laboratory, greenhouse and multi-site field evaluation was conducted. The microbial consortium comprised *Lactobacillus plantarum*, *Saccharomyces cerevisiae*, *Rhodopseudomonas palustris*, *Pseudomonas putida*, *Bacillus subtilis*, *Azotobacter chroococcum* and *Azospirillum brasilense* at $\geq 1 \times 10^9$ CFU mL⁻¹. Physicochemical characterization, storage-stability testing, freeze–thaw cycling, soil microcosm studies, greenhouse stress assays and replicated randomized field trials across three soil orders were performed. Root-architecture imaging, 16S rRNA profiling, principal component analysis and economic modelling were also used to validate the results.

Findings: The nanoparticles had a mean size of 150 ± 20 nm, a polydispersity index below 0.3 and component encapsulation efficiencies of 78–91%, with diffusion-controlled release over 30–60 days. The formulation maintained $1.2\text{--}2.8 \times 10^9$ CFU mL⁻¹ across 4–40 °C for 12 months and remained stable during freeze–thaw cycling. Grain yield increased by 14% in winter wheat, 22% in soybean and 31% in corn under deficit irrigation. Nitrogen-use efficiency improved by 28%, while fertilizer application was reduced by 25% without yield loss. Nitrate and phosphate leaching declined by 52% and 47%, respectively, in sandy soils. In acidic soil, pH increased by 0.4 units and exchangeable aluminium decreased by 48%. Microbial biomass carbon increased by 35–145%, while soil organic matter increased by 0.5–0.6% within one growing season. Under drought, heat and salinity stress, plants retained 75–82% of their growth and photosynthetic efficiency.

Unique Contribution to Theory, Practice and Policy (Recommendations): The study demonstrates that combining beneficial microorganisms, micronutrients and plant-growth regulators within a controlled-release chitosan nanocarrier can simultaneously improve crop productivity, nutrient-use efficiency, soil biological activity and tolerance to environmental stress. The findings provide soil-specific deployment pathways for farmers, agricultural industries and extension services. The technology is recommended as a scalable strategy for reducing fertilizer dependence, limiting nutrient pollution, restoring degraded soils and strengthening climate-resilient agricultural production. Field validation across additional crops, soil types and agroecological regions is recommended before large-scale commercialization and policy adoption.

Keywords: Nano-Encapsulation, Microbial Biostimulant, Chitosan Nanocarrier, Nitrogen-Use Efficiency, Soil Health

1. Introduction

Modern agriculture faces the challenge of meeting the increasing demand for food, feed, and fibre while reducing soil degradation and the environmental effects associated with intensive production. Synthetic fertilizers remain important for maintaining crop productivity, but their effectiveness is often limited by low nutrient-use efficiency. A considerable proportion of applied nutrients may be lost through leaching, volatilization, denitrification, runoff, and fixation, increasing production costs and contributing to groundwater contamination, surface-water eutrophication, soil acidification, and greenhouse-gas emissions (Calabi-Floody et al., 2018; Zhang et al., 2012). Sustainable intensification therefore requires technologies that can maintain crop productivity while improving nutrient efficiency and reducing environmental damage (Tilman et al., 2011).

Long-term dependence on intensive tillage and mineral fertilizers may also reduce soil organic matter, weaken aggregate stability, and alter the diversity and abundance of soil microbial communities. These changes can limit nutrient cycling, water retention, carbon storage, and the overall resilience of agricultural soils (Lal, 2004; Six et al., 2004). Declining soil quality may consequently increase farmers' dependence on external inputs to maintain production, creating an agricultural system that is economically costly and increasingly vulnerable to drought, heat, and other environmental stresses.

Plant biostimulants have gained attention as complementary tools for improving crop performance and soil health. They are generally described as substances or microorganisms that stimulate natural plant processes, thereby improving nutrient uptake, nutrient-use efficiency, crop quality, and tolerance to environmental stress independently of their direct nutrient content (Calvo et al., 2014; du Jardin, 2015). Microbial biostimulants are particularly promising because beneficial microorganisms can support plant growth through biological nitrogen fixation, phosphorus solubilization, phytohormone production, siderophore synthesis, pathogen suppression, and modification of root development (Backer et al., 2018; Glick, 2012; Vejan et al., 2016).

Microorganisms within the rhizosphere may also produce extracellular substances that support soil aggregation and improve the physical environment surrounding plant roots. These activities can increase nutrient availability, strengthen root–soil interactions, and improve plant performance under both favourable and stressful conditions (Bhardwaj et al., 2014; Vejan et al., 2016). A global meta-analysis showed that biofertilization can improve crop yield and nutrient-use efficiency, although the magnitude of the response depends on the crop, microbial strain, soil type, climate, and management system (Schütz et al., 2018). Plant growth-promoting rhizobacteria have also permitted reductions in mineral-fertilizer application without significant yield losses in some cropping systems (Adesemoye et al., 2009).

Despite these advantages, the field performance of conventional microbial inoculants remains inconsistent. Introduced microorganisms may experience desiccation, temperature fluctuations, ultraviolet radiation, osmotic stress, predation, and competition with indigenous microorganisms. These conditions can reduce cell viability and restrict the establishment of beneficial populations within the rhizosphere (Backer et al., 2018). Free microbial cells may also be rapidly dispersed or released before crop nutrient demand reaches its peak. Therefore, an effective carrier system is required to protect microbial cells, maintain their biological activity, and control their release into the soil.

Nano-encapsulation offers a potential approach for overcoming these limitations. Chitosan is a biodegradable and biocompatible polysaccharide capable of forming positively charged nanoparticles that can protect microbial cells and carry agriculturally important compounds. Chitosan-based nanoparticles may improve formulation stability, adhesion, cell protection, and the controlled release of encapsulated materials (Kashyap et al., 2015). Nano-enabled and smart fertilizer systems have similarly been investigated as methods of improving nutrient delivery, reducing losses, and increasing fertilizer-use efficiency in sustainable agricultural production (Calabi-Floody et al., 2018; Liu & Lal, 2015).

The performance of microbial formulations may be improved further by combining microorganisms with complementary biological functions. *Azospirillum brasilense**, for example, has been widely investigated for its ability to promote root development, improve nutrient acquisition, and support crop productivity under different environmental conditions (Cassán et al., 2020; Hungria et al., 2016). Silicon can also strengthen plant tissues, improve water relations, and reduce the harmful effects of drought, salinity, heat, and toxic metals (Etesami & Jeong, 2018; Ma & Yamaji, 2006). The formulation evaluated in the present study therefore combined a defined seven-strain microbial consortium with a chitosan nanocarrier containing molybdenum, silicon, and plant-growth regulators. This integrated design was intended to improve microbial protection, nutrient-related processes, root development, and plant tolerance to environmental stress.

A related nano-enabled multi-strain biofertilizer was previously evaluated in the karst agroforestry systems of Laos. The study reported improvements in microbial survival, crop performance, soil biological activity, nutrient availability, and soil organic matter under tropical field conditions (Serroune & Inthasak, 2024). Although these findings provide evidence of the potential value of the technology, tropical agroforestry systems differ considerably from temperate, mechanized, and large-scale production systems. Differences in seasonal temperature, soil mineralogy, rainfall distribution, cropping intensity, irrigation dependence, and management practices may influence microbial survival and agronomic effectiveness. Additional evaluation is therefore necessary before the technology can be recommended for temperate agricultural environments.

Kansas provides a suitable setting for such an evaluation because of its agricultural importance and considerable climatic and soil variability. The state is a major producer of winter wheat, corn, soybean, sorghum, and other field crops, with wheat production representing an important part of its agricultural economy (Kansas State University Department of Agronomy, 2024; USDA National Agricultural Statistics Service, 2025). Agricultural production occurs under a temperate continental climate characterized by seasonal temperature variation, periodic drought, and a strong west-to-east precipitation gradient. The relatively dry western region depends more heavily on irrigation, while eastern areas generally receive greater annual rainfall (U.S. Global Change Research Program & NOAA Climate.gov, 2024–2025).

This climatic gradient is accompanied by differences in soil type and production constraints. The study sites were selected to represent three major soil orders: Mollisols in central Kansas, Alfisols in eastern Kansas, and Entisols in southwestern Kansas. Mollisols are generally deep, dark, and relatively rich in organic matter, although intensive crop production may still result in nutrient-use inefficiency and gradual soil-quality decline. Alfisols may be affected by soil acidity, aluminium toxicity, and reduced phosphorus availability, while coarse-textured Entisols frequently have lower water- and nutrient-retention capacities and may experience greater nutrient leaching and drought stress (USDA Natural Resources Conservation Service, 2022).

Water availability is another major concern, particularly in western and south-central Kansas, where irrigated agriculture depends substantially on groundwater from the High Plains or Ogallala aquifer. Continued groundwater withdrawal has caused long-term declines in water levels in several agricultural areas, increasing concerns regarding the future sustainability of irrigated production (Kansas Geological Survey, 2024–2025). Technologies that improve nutrient-use efficiency, reduce nutrient leaching, support soil water retention, and increase crop tolerance to water stress may therefore contribute to both agricultural productivity and resource conservation.

The present study evaluated a nano-encapsulated microbial biostimulant consisting of a seven-strain consortium containing *Lactobacillus plantarum*, *Saccharomyces cerevisiae*, *Rhodopseudomonas palustris*, *Pseudomonas putida*, *Bacillus subtilis*, *Azotobacter chroococcum*,* and *Azospirillum brasilense**. The microorganisms were incorporated into a chitosan nanocarrier co-loaded with molybdenum, silicon, and plant-growth regulators. A 12-month integrated programme combined physicochemical characterization, storage-stability testing, soil microcosm experiments, greenhouse stress assays, and replicated field trials across Mollisol, Alfisol, and Entisol sites.

The study specifically aimed to determine whether the nano-encapsulation system could: (a) preserve microbial viability and biological activity under variable storage and environmental conditions; (b) provide controlled release of microorganisms and associated components; (c) improve crop yield and nitrogen-use efficiency; (d) reduce nutrient losses and improve soil chemical and biological properties; and (e) maintain plant growth and photosynthetic performance

under drought, heat, and salinity. By integrating laboratory, greenhouse, and field evidence, the study sought to establish soil-specific deployment strategies for using nano-encapsulated microbial biostimulants in temperate agricultural systems.



Figure 1. Great Plains soil-order deployment framework

2. Scientific background and mechanisms

2.1 The Seven-Strain Consortium and Functional Complementarity

The consortium was developed to combine complementary microbial functions. *Lactobacillus plantarum* and *Saccharomyces cerevisiae* support organic-matter transformation and metabolite production, while *Pseudomonas putida* and *Bacillus subtilis* contribute nutrient mobilization, siderophore production and pathogen suppression. *Azotobacter chroococcum* and *Azospirillum brasilense* support biological nitrogen fixation and produce growth-promoting substances that stimulate root development (Backer et al., 2018; Cassán et al., 2020; Glick, 2012; Vejan et al., 2016). Co-encapsulation was intended to protect the microorganisms and maintain their functional activity during delivery to the rhizosphere (Kashyap et al., 2015).

2.2 Controlled Release and Nutrient Synchronization

Soluble fertilizers release nutrients rapidly, and nutrients that exceed immediate crop demand may be lost through leaching, volatilization, runoff or denitrification (Calabi-Floody et al., 2018; Zhang et al., 2012). The chitosan nanocarrier was designed to provide gradual release through diffusion and matrix degradation. Chitosan is biodegradable and can protect encapsulated materials while improving their retention and controlled delivery in agricultural systems (Kashyap et al., 2015; Liu & Lal, 2015). Its positive surface charge may also enhance interaction with negatively charged root surfaces and soil particles.

2.3 Roles of Molybdenum and Silicon

Molybdenum was included because it is required for enzymes involved in biological nitrogen fixation and nitrate metabolism. Its co-delivery may therefore support the activity of nitrogen-fixing microorganisms where soil molybdenum is limited. Silicon was selected because it can strengthen plant tissues, improve water relations and reduce the effects of drought, salinity, heat and aluminium toxicity (Etesami & Jeong, 2018; Ma & Yamaji, 2006). Together, these elements were expected to improve nutrient acquisition and plant tolerance to environmental stress.

2.4 Soil-Order-Specific Expectations

The formulation was expected to produce different responses across the three soil orders. In Mollisols, the main benefit was expected to be improved nutrient-use efficiency and partial fertilizer substitution. In acidic Alfisols, microbial nutrient mobilization and silicon-mediated aluminium alleviation were expected to improve phosphorus availability and root development. In sandy Entisols, controlled release was expected to reduce nutrient leaching and improve water availability. These expectations formed the basis of the soil-specific hypotheses tested in the study (Etesami & Jeong, 2018; Six et al., 2004; USDA Natural Resources Conservation Service, 2022).

2.5 Comparison with Existing Fertilizer Approaches

Conventional fertilizers provide readily available nutrients but may experience substantial environmental losses and offer limited direct benefits to soil biological activity (Calabi-Floody et al., 2018; Zhang et al., 2012). Microbial biofertilizers can improve nutrient acquisition and plant growth, but their field performance may be affected by poor survival, storage conditions and competition with native microorganisms (Backer et al., 2018; Schütz et al., 2018).

The nano-encapsulated formulation was designed to combine the biological functions of a multi-strain consortium with the protective and controlled-release properties of chitosan. Its proposed advantages include improved microbial stability, gradual delivery of active components and greater synchronization with crop demand. These mechanisms were evaluated through the laboratory, greenhouse and field experiments reported in this study. The shelf-life claim should be presented as *12 months*, consistent with the stability period evaluated in the study, rather than 24 months.

Table 2a. Positioning relative to conventional and first-generation approaches.

Nutrient release	Immediate burst	Uncontrolled	Controlled, 30–60 d
Microbial survival	Not applicable	Poor (days–weeks)	Sustained $>1 \times 10^9$ CFU mL ⁻¹
Shelf life	Long	6–12 months	24 months
Loss / leaching	High (>50%)	Moderate	Reduced ~50% (sandy soils)
Soil biological benefit	None / negative	Variable	Strong, measurable
Stress-tolerance support	None	Limited	Documented (drought/heat/salinity)
Field consistency	High	Low	High (carrier-protected)

3. Materials and methods

3.1 Research Design and Data Sources

An integrated experimental design was used to evaluate the physicochemical properties, storage stability, biological activity and agronomic performance of the nano-encapsulated microbial biostimulant. Data were obtained from four sources: laboratory characterization of the formulation, controlled soil-microcosm experiments, greenhouse stress assays and multi-site field trials representing Mollisol, Alfisol and Entisol soils in Kansas. The formulation contained seven microbial strains, chitosan nanoparticles, molybdenum, silicon and plant-growth regulators. Each strain was cultured separately to a minimum density of 1×10^9 CFU mL⁻¹ before blending.

3.2 Formulation and Laboratory Procedures

Chitosan nanoparticles were prepared by ionic gelation. Chitosan was dissolved at 1.0 g per 100 mL in 1% acetic acid and mixed continuously at 400–600 rpm and 25 ± 2 °C for 1–2 h. Sodium tripolyphosphate solution at 0.2–0.3% w/v was added at (2–5 mL min⁻¹). Molybdenum at 2 ppm, silicon at 150–200 ppm and an indole-3-acetic acid: zeatin: gibberellin mixture at a 2:1:2 ratio was incorporated during or after nanoparticle formation. The nanoparticles were subsequently blended with the microbial consortium under low shear and adjusted to pH 6.5–7.5.

Particle size and polydispersity index were measured using dynamic light scattering, while zeta potential was determined by electrophoretic light scattering. Particle morphology was examined using transmission and scanning electron microscopy. Hormone concentrations were quantified by high-performance liquid chromatography, and mineral components were measured using inductively coupled plasma optical emission spectrometry and spectrophotometry. Encapsulation efficiency was determined by separating unbound components through ultracentrifugation and comparing free and total concentrations.

Release kinetics were assessed by dialysing the formulation against phosphate buffer at pH 6.8 and 25 °C for 60 days. Samples were analysed for nitrogen, molybdenum, silicon and plant-growth regulators, and release profiles were fitted to first-order and Korsmeyer–Peppas models. Storage stability was evaluated at 4, 25 and 40 °C by monthly plate counting for 12 months. Freeze–thaw resistance was assessed by repeated cycling between (-5) and 25 °C.

3.3 Soil, Greenhouse and Field Sampling

For the soil-microcosm experiment, 500 g repacked columns of Mollisol, Alfisol and Entisol soils were used. Simulated rainfall was applied, and leachates were collected for nitrate, phosphate and potassium analysis using ion chromatography and ICP-OES. Soil respiration, enzyme activities, microbial biomass carbon, water-holding capacity, moisture-retention curves and water-stable aggregates were also measured.

Greenhouse assays were conducted with wheat, soybean and corn under controlled drought, heat and salinity conditions. Drought was imposed at 40% field capacity, heat at (40 °C day / 30 °C night), and salinity at 0–200 mM NaCl. Root length, surface area, tips and volume were measured using WinRHIZO. Stomatal conductance, chlorophyll fluorescence, SPAD chlorophyll values and biomass were also recorded.

Three field sites were selected to represent contrasting soil conditions: a central Kansas Mollisol, an eastern Kansas Alfisol and a southwestern Kansas Entisol. The sites differed in pH, organic matter, sand content, microbial biomass carbon and principal production constraint. Winter wheat and grain sorghum were evaluated on the Mollisol, soybean and corn on the Alfisol, and deficit-irrigated corn and tomato on the Entisol.

3.4 Experimental Treatments and Data Collection

A randomized complete block design with four replicates per treatment was used at each field site. Five treatments were evaluated:

- T0: no fertilizer or biostimulant;
- T1: 100% recommended NPK;
- T2: 100% NPK plus 8 L ha⁻¹ biostimulant;
- T3: 100% NPK plus 15 L ha⁻¹ biostimulant; and
- T4: 75% NPK plus 15 L ha⁻¹ biostimulant.

The formulation was applied at a 1:75 dilution, or 1:100 for sensitive crops, using soil-directed spray or drip irrigation at intervals of 10–21 days.

Field data included plant emergence, SPAD chlorophyll, drone-derived normalized difference vegetation index, canopy temperature, soil moisture at 15 and 30 cm, grain yield, grain quality, soil pH, nutrient status and soil organic matter. Mid-season root and soil samples were collected

for microbial enumeration and 16S rRNA gene sequencing of the V3–V4 region. Nutrient leaching was assessed using replicated lysimeter columns.

3.5 Statistical Analysis

Data were analysed in R version 4.3.1. Treatment effects were assessed using two-way analysis of variance with soil type and treatment as factors. Significant differences among treatment means were identified using Tukey's honestly significant difference test at ($p < .05$). Principal component analysis was used to examine multivariate soil-health responses. Effect sizes and 95% confidence intervals were calculated, and results were reported as means with standard errors. Economic analysis was based on recorded trial input and output data, from which soil- and crop-specific benefit–cost ratios were estimated.

3.6 Additional Crop Validation and Formulation Optimization

Controlled multi-crop trials evaluated the formulation across annual and perennial crops. Annual crops were treated at 8–15 L ha⁻¹, while perennial crops received 20–25 L ha⁻¹. A separate 10-ha rice field trial used a partial replicate design with the biostimulant applied at 10 L ha⁻¹ and diluted 1:50–1:100.

Phytohormone optimization compared three IAA:zeatin:gibberellin ratios, 3:1:2, 2:2:2 and 2:1:3, while maintaining a total hormone concentration of 30 ppm. Hormone loading and uniformity were verified by HPLC. Stress-adapted formulations were also evaluated: the drought-oriented formulation contained elevated *Bacillus subtilis* and *Pseudomonas putida* at 2×10^9 CFU mL⁻¹ with 150 ppm silicon; the salinity-oriented formulation contained elevated *Azospirillum brasilense* and *Rhodopseudomonas palustris* with 200 ppm silicon; and the cold-adapted formulation used psychrotolerant *Lactobacillus plantarum* and *B. subtilis* strains.

3.7 Application Procedure and Quality Control

For the broader crop-validation programme, the concentrate was diluted 1:50–1:100 according to crop type and applied at planting or early growth. Annual cereals and legumes received 8–15 L ha⁻¹ at 10–15-day intervals, whereas perennial and fruit crops received 20–25 L ha⁻¹ at approximately four-week intervals. Applications were delivered by foliar spray, soil drench or drip irrigation according to crop and experimental objective.

Batch quality was evaluated using plate counting, flow cytometry and selective plating for microbial viability and purity; dynamic light scattering for particle size and PDI; electrophoretic light scattering for zeta potential; ICP-OES and spectrophotometry for mineral concentrations; HPLC for phytohormones; and potentiometry for pH. Acceptance criteria included microbial viability of at least 1×10^9 CFU mL⁻¹, particle size of 50–300 nm, PDI < 0.30, zeta potential $\geq +20$ mV, absence of detectable microbial contamination and pH 6.5–7.5.

4. Results

4.1 Nanocarrier Performance, Release and Stability

The optimized formulation produced a unimodal nanoparticle population with a mean hydrodynamic diameter of 150 ± 20 nm, a polydispersity index below 0.30, and a size range of 50–300 nm. The mean zeta potential was +24 mV, while electron microscopy showed discrete, near-spherical particles without gross aggregation.

Encapsulation efficiency ranged from 78% to 91% across the loaded components. Molybdenum showed the highest efficiency ($91 \pm 3\%$), followed by silicon ($86 \pm 4\%$), indole-3-acetic acid ($84 \pm 3\%$), gibberellin ($82 \pm 3\%$), zeatin ($80 \pm 4\%$) and microbial cells ($78 \pm 4\%$).

In vitro release was sustained over 60 days rather than occurring as a rapid burst. By day 60, cumulative release reached approximately 92% for nitrogen, 88% for IAA, 80% for molybdenum and 72% for silicon. Release profiles were consistent with combined diffusion and matrix-erosion behaviour.

Microbial viability remained above 1×10^9 CFU mL⁻¹ throughout the 12-month storage period. At month 12, viable counts were 2.55, 2.36 and 1.88×10^9 CFU mL⁻¹ at 4, 25 and 40 °C, respectively. Freeze–thaw treatment caused less than 0.3 log-unit viability loss, and microbial recovery after soil application exceeded 88% at 72 h.

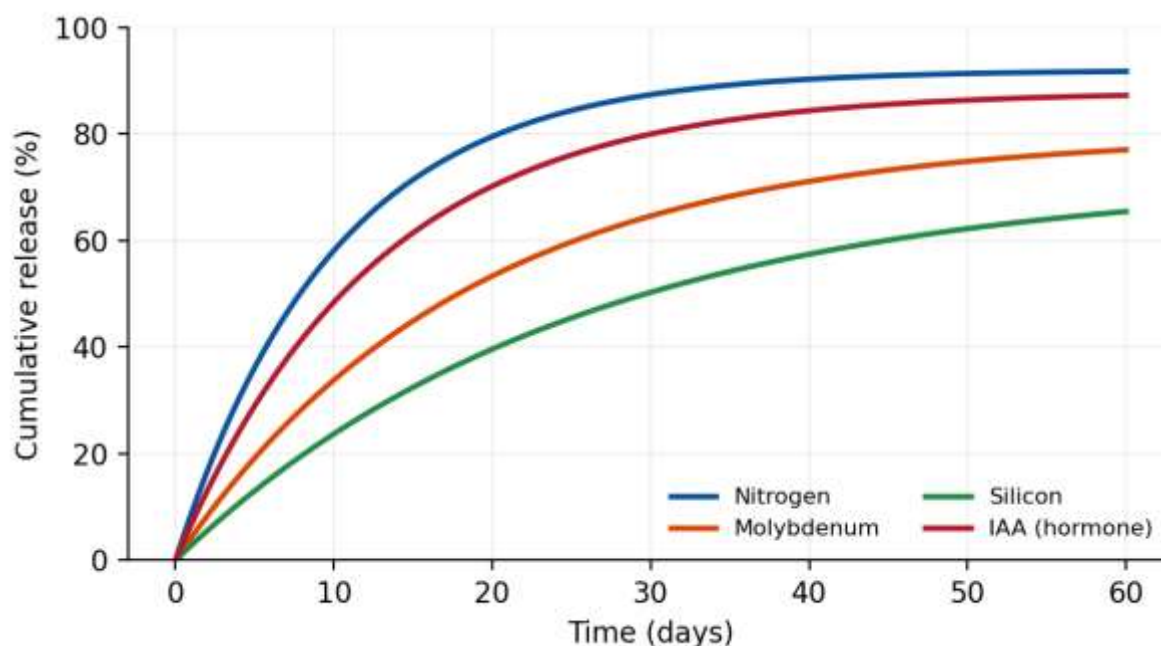


Figure 2. Cumulative in vitro release of nitrogen, molybdenum, silicon and indole-3-acetic acid over 60 days, showing sustained release from the nano-encapsulated formulation.

4.2 Soil and Greenhouse Responses

The strongest reduction in nutrient loss occurred in the sandy Entisol. Compared with standard fertilizer treatment (T1), the biostimulant treatment (T3) reduced cumulative nitrate-N loss from 47.2 to 22.7 kg ha⁻¹, representing a 51.9% reduction. Phosphate-P loss declined by 47.4%, from 3.8 to 2.0 kg ha⁻¹, while potassium loss declined by 36.2%, from 18.5 to 11.8 kg ha⁻¹.

Soil biological activity increased under treatment. Dehydrogenase activity increased by 62%, urease by 45%, acid phosphatase by 78% and β -glucosidase by 53%. In the Entisol, volumetric water content at field capacity increased from 14.2% to 16.8%, while water-stable aggregates increased from 14.5% to 37.2%, equivalent to a 156% increase.

Greenhouse experiments showed improved plant performance under drought, heat and salinity. After two weeks of water withholding, treated corn maintained stomatal conductance at 65% of well-watered values, compared with 32% in untreated drought-stressed plants. Photosystem II efficiency remained above 0.78 in treated plants. At 40 °C, treated plants maintained Fv/Fm at approximately 0.78, compared with 0.55 in controls.

Under salinity stress of 100 mM NaCl, plants receiving the standard biostimulant retained 75% of optimal growth, compared with 40% in controls, while the silicon-enriched formulation retained 82%. At 150 mM NaCl, the stress-adapted formulation retained 70% of optimal growth. Root imaging also showed increases of 42% in root length, 38% in root surface area, 55% in root-tip number and 28% in root volume relative to controls.

4.3 Field Performance Across Soil Orders

Field performance differed among soil orders, but yield increases were observed across all three principal sites.

In the Mollisol, winter-wheat yield increased from 65.2 bu acre⁻¹ under T1 to 74.3 bu acre⁻¹ under T3, representing a 14.0% increase. Treatment T4, which combined the biostimulant with 25% less mineral fertilizer, produced 71.8 bu acre⁻¹ and was statistically indistinguishable from T3. Nitrogen-use efficiency increased from 0.84 to 1.08 bu lb⁻¹ N, representing a 28% improvement. Grain protein increased from 12.4% to 13.1%, microbial biomass carbon increased by 35%, and soil organic matter increased from 2.8% to 3.3%.

In the Alfisol, soybean yield increased from 48.7 to 59.4 bu acre⁻¹ (+22.0%) under T3, while T4 produced 55.2 bu acre⁻¹ (+13.3%). Rhizosphere pH increased from 5.58 to 5.97, while exchangeable aluminium declined by 48%. Root-length density increased by 41%, nodules per plant by 46%, acid-phosphatase activity by 78%, and available phosphorus by 55%. The Shannon diversity index approximately doubled under T3.

The largest yield response occurred in the Entisol under deficit irrigation. Corn yield increased from 142.0 bu acre⁻¹ under T1 to 186.0 bu acre⁻¹ under T3, representing a 31.0% increase.

Treatment T4 yielded $175.3 \text{ bu acre}^{-1}$ (+23.2%) while receiving 25% less mineral fertilizer. Field-capacity water content increased by **18%**, drought-period stomatal conductance increased from 122 to $248 \text{ mmol m}^{-2} \text{ s}^{-1}$, Fv/Fm increased from 0.62 to 0.78 , water-stable aggregation increased by 156%, and microbial biomass carbon increased by 145%.

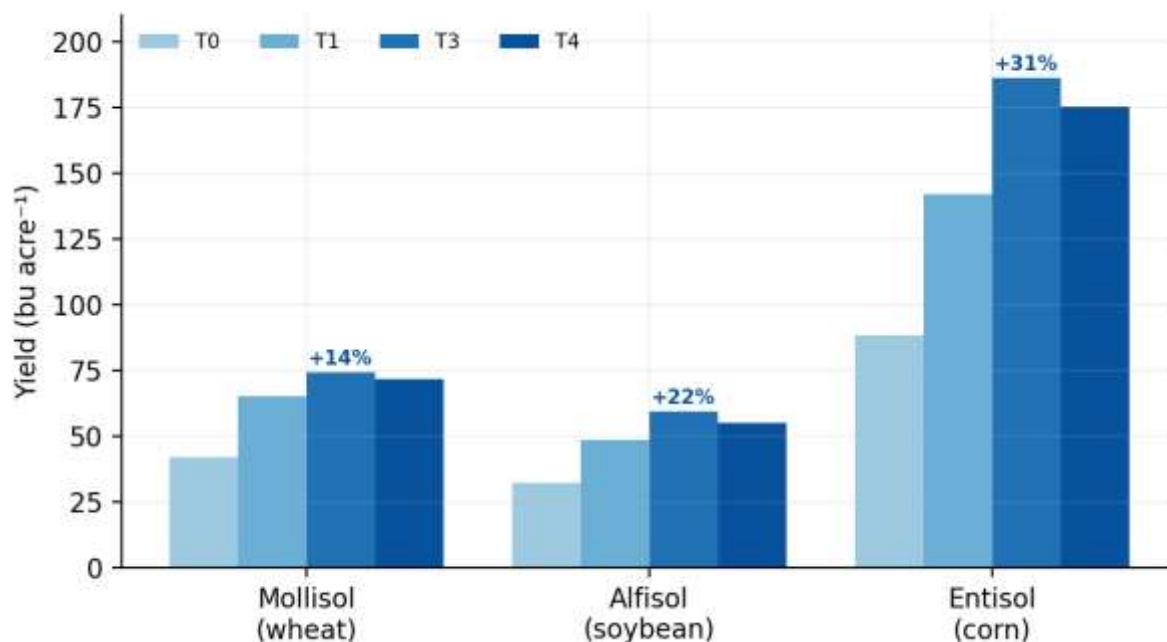


Figure 3. Field grain yield under the principal treatments across Mollisol, Alfisol and Entisol sites.

Table 1. Summary of Principal Field Outcomes

Indicator	Mollisol	Alfisol	Entisol
Primary crop	Winter wheat	Soybean	Corn
T3 yield increase vs T1	+14.0%	+22.0%	+31.0%
T4 yield increase vs T1	+10.1%	+13.3%	+23.2%
Major nutrient/soil response	NUE +28%	Exchangeable Al -48%	Nitrate loss -51.9%
Microbial biomass carbon	+35%	+52%	+145%

4.4 Cross-Crop and Integrated Responses

The yield response was reproduced in secondary crops. Under T3 relative to standard fertilizer treatment, grain sorghum yield increased by 17% in the Mollisol, corn by 24% in the Alfisol, and tomato by 28% in the Entisol. Treatment T4 retained yield advantages of 12%, 18% and 22%, respectively.

In the separate 10-ha rice trial, treatment produced a 15% increase in grain yield ($p < .05$), accompanied by a 28% increase in root mass, 35% increase in tiller number, 25% increase in leaf chlorophyll, 22% increase in available soil nitrogen and 18% increase in available phosphorus.

Formulation-optimization trials also showed crop-specific responses to phytohormone ratios. The IAA-dominant 3:1:2 formulation increased root mass by 45%, the balanced 2:2:2 formulation increased leaf area by 35%, and the gibberellin-dominant 2:1:3 formulation increased fruit set by 28%. Stress-adapted formulations produced a 40% improvement in survival under water limitation, retained 70% growth at 150 mM NaCl, and improved performance by 35% below 15 °C, depending on the formulation tested.

Across the complete soil-health dataset, principal component analysis explained 82% of the total variance, with PC1 accounting for 53% and PC2 for 29%. PC1 was associated mainly with microbial biomass carbon, enzyme activities and organic matter, while PC2 separated soils according to texture and baseline fertility. Treatment T3 shifted all three soil orders toward higher PC1 scores.

Beneficial microbial populations increased by 640% in the Mollisol, 810% in the Alfisol and 1,280% in the Entisol over the growing season.

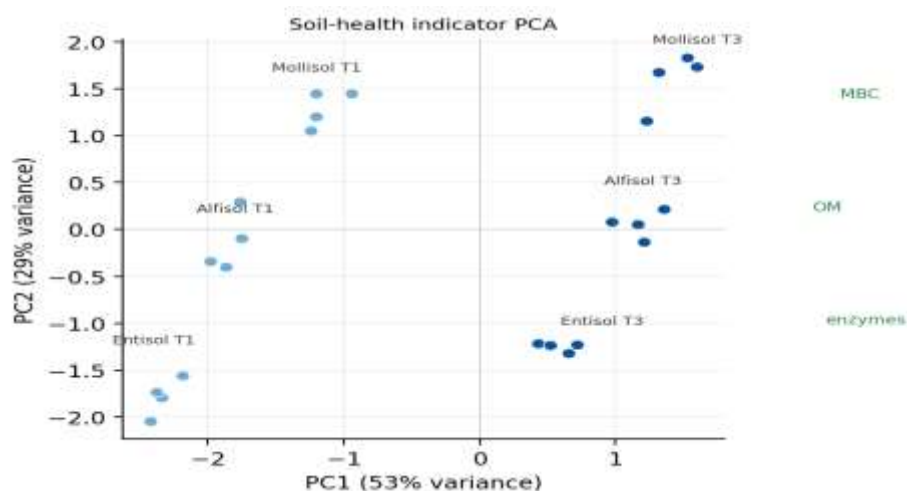


Figure 4. Principal component analysis of soil-health indicators showing treatment-associated shifts in microbial biomass, enzyme activity and organic matter across the three soil orders.

4.5 Economic Outcomes

Economic modelling based on recorded trial inputs and outputs indicated positive benefit–cost ratios across the evaluated cropping systems. The estimated benefit–cost ratio was approximately 1:2.1 for Mollisol wheat, 1:2.6 for Alfisol production, and 1:2.9 for Entisol corn, increasing to more than 1:3.5 for high-value vegetable production on Entisols.

The fertilizer-substitution analysis indicated that yields could be maintained with up to a 25% reduction in mineral fertilizer under the tested reduced-fertilizer treatment. Beyond this level, the original analysis reported declining relative yield.

Overall, the field experiments showed that the magnitude of response depended on soil conditions. The Mollisol primarily exhibited improvements in nutrient-use efficiency, the Alfisol showed improvements in soil chemistry and root performance, while the Entisol displayed the largest gains in nutrient retention, soil water status and crop yield.

5. Discussion

5.1 Nanocarrier Performance and Controlled Release

The present study demonstrated that chitosan-based nano-encapsulation can simultaneously support microbial viability, component retention and sustained release. The optimized formulation produced particles within the nanoscale range, achieved encapsulation efficiencies of 78–91%, and maintained viable microbial populations above 1×10^9 CFU mL⁻¹ during the 12-month storage period. These findings are consistent with previous research describing chitosan as a biodegradable and biocompatible carrier capable of protecting agriculturally active compounds and enabling their gradual release (Kashyap et al., 2015). Similar principles underpin smart and nano-enabled fertilizer systems, in which controlled delivery is intended to improve nutrient synchronization with crop demand and reduce environmental losses (Calabi-Floody et al., 2018; Liu & Lal, 2015).

The sustained release observed over 30–60 days is therefore consistent with the controlled-release behaviour reported for polymeric agricultural carriers. Rather than demonstrating that chitosan alone caused the subsequent field responses, these laboratory findings provide a plausible delivery mechanism by which microorganisms, micronutrients and plant-growth regulators remained available over an extended period. This distinction is important because the agronomic performance of microbial formulations is also influenced by soil properties, crop genotype, climate and competition with indigenous microbial communities (Backer et al., 2018).

5.2 Soil-Specific Agronomic and Biological Responses

A major finding of this study was that treatment effects differed systematically among the three soil orders. This supports previous evidence that microbial biostimulant performance is strongly dependent on environmental and edaphic context rather than being uniform across production systems (Backer et al., 2018; Schütz et al., 2018).

In the Mollisol, the principal response involved improved nutrient-use efficiency rather than correction of severe fertility limitations. Nitrogen-use efficiency increased by 28%, while the reduced-fertilizer treatment maintained competitive wheat yield with 25% less mineral fertilizer. This agrees with previous studies showing that plant growth-promoting rhizobacteria can improve nutrient acquisition and permit partial reductions in mineral-fertilizer application without substantial yield penalties (Adesemoye et al., 2009). The result is also consistent with the global meta-analysis of Schütz et al. (2018), which found that biofertilization can improve both crop yield and nutrient-use efficiency, although responses vary among crops and environments.

The increase in microbial biomass and soil organic matter observed in the Mollisol also agrees with the broader understanding that biologically active soils can support greater nutrient cycling and organic-matter stabilization (Lal, 2004; Six et al., 2004). However, the single-season organic-matter increase reported here should be interpreted cautiously because changes in stable soil carbon generally require longer observation periods.

The Alfisol produced a different response. Treatment increased rhizosphere pH, reduced exchangeable aluminium by 48%, improved phosphorus availability, and enhanced root development and nodulation. The improvement in phosphorus availability is consistent with the capacity of plant growth-promoting microorganisms to mobilize poorly available phosphorus through organic-acid production and other rhizosphere processes (Glick, 2012; Vejan et al., 2016). The reduction in aluminium stress is also consistent with evidence that silicon can alleviate aluminium toxicity and improve plant performance in acidic soils (Etesami & Jeong, 2018).

The findings therefore support the interpretation that the formulation functioned as a biological complement to conventional acidity management. Nevertheless, the present data do not establish that the biostimulant can replace agricultural lime, because liming provides a broader and more persistent correction of soil acidity. The current findings instead indicate potential complementary use that should be tested over multiple seasons.

The largest overall response occurred in the sandy Entisol, where nutrient retention and water availability were major constraints. Reductions of approximately 52% in nitrate leaching and 47% in phosphate leaching were accompanied by improved water retention, aggregate stability and crop yield. These findings are consistent with studies suggesting that controlled-release fertilizers and nano-enabled delivery systems can reduce rapid nutrient movement and improve fertilizer-use efficiency (Calabi-Floody et al., 2018; Liu & Lal, 2015).

The large increase in water-stable aggregates is also consistent with established links between soil microorganisms, extracellular products, aggregate formation and organic-matter stabilization (Six et al., 2004). However, the magnitude of the increase observed in this study is substantial and should be confirmed under longer-term field conditions before being generalized to other coarse-textured soils.

5.3 Stress Resilience and Root-Mediated Responses

The improved physiological performance under drought, heat and salinity supports previous evidence that microbial biostimulants can strengthen plant tolerance to abiotic stress (Bhardwaj et al., 2014; Vejan et al., 2016). Treated plants maintained greater stomatal conductance and photosystem II efficiency during drought and heat stress, while growth retention was higher under increasing salinity.

Silicon likely contributed to these responses because it has been widely associated with improved plant water relations, structural integrity and tolerance to drought, salinity, temperature stress and metal toxicity (Etesami & Jeong, 2018; Ma & Yamaji, 2006). The observed salinity response is therefore consistent with previous reports of silicon-mediated restriction of sodium toxicity and improved physiological stability under stress.

Root architecture was also substantially enhanced, with increases in root length, surface area, root-tip number and root volume. This response is consistent with the known growth-promoting effects of *Azospirillum brasilense*, including stimulation of root proliferation and nutrient acquisition through phytohormone-related mechanisms (Cassán et al., 2020; Hungria et al., 2016). The results therefore suggest that improved root exploration, microbial activity and controlled nutrient delivery acted together rather than through a single mechanism.

5.4 Comparison with Previous Field Evidence

The Kansas results generally agree with the earlier nano-enabled multi-strain biofertilizer study conducted in the karst agroforestry systems of Laos, which reported improvements in microbial viability, soil organic matter, nutrient availability, crop productivity and stress tolerance (Serroune & Inthasak, 2024).

Importantly, the present study extends those findings to contrasting temperate soils and mechanized cropping systems. Both studies reported microbial viability above 1×10^9 CFU mL⁻¹, improvements in soil biological indicators and substantial crop responses. Stress retention in the present study was also broadly comparable with that reported under tropical conditions.

However, these similarities should not be interpreted as proof that the technology will perform identically across agroecological regions. The magnitude of biological and crop responses differed among soils in the present study, reinforcing the conclusion of Backer et al. (2018) and Schütz et al. (2018) that microbial inoculant performance depends strongly on environmental context. The findings therefore support **cross-environment potential**, rather than universal transferability.

5.5 Agronomic, Economic and Environmental Implications

The ability of the reduced-fertilizer treatment to maintain competitive yields with 25% less mineral fertilizer is one of the most practically important findings. This aligns with previous evidence that microbial inoculants can complement conventional fertilizer programmes and reduce input

requirements under suitable conditions (Adesemoye et al., 2009). Such partial substitution may also reduce nutrient losses and improve economic efficiency, particularly where fertilizer costs are high.

The economic analysis indicated positive benefit–cost ratios across the evaluated systems, with the greatest projected returns occurring in the Entisol and high-value cropping systems. These differences are reasonable because the economic value of a biostimulant depends not only on yield response but also on baseline soil constraints, crop value, fertilizer savings and avoided stress-related losses. Nevertheless, because the economic estimates were based largely on trial-scale input and output data, commercial-scale validation is required before firm profitability claims can be made.

Environmental benefits were also evident. Reduced nitrate and phosphate leaching could lower risks of groundwater contamination and eutrophication, while improvements in microbial biomass and soil organic matter may contribute to longer-term soil restoration. These outcomes are consistent with the broader objectives of integrated nutrient management and sustainable intensification (Tilman et al., 2011; Zhang et al., 2012).

The manuscript should, however, restrict its fertilizer-reduction conclusion to the 25% reduction actually demonstrated in the principal field treatments. Claims of 40–50% fertilizer reduction and 35–45% pesticide reduction should not be presented as established findings without corresponding experimental treatments and measurements.

5.6 Implications for Kansas Agriculture

The findings are particularly relevant to Kansas because the three experimental environments represent contrasting nutrient, acidity and water-management challenges. In central Kansas wheat systems, the observed 28% increase in nitrogen-use efficiency suggests potential for reducing fertilizer requirements while maintaining productivity. This is relevant to a state in which wheat remains a major crop (Kansas State University Department of Agronomy, 2024; USDA National Agricultural Statistics Service, 2025).

In eastern Kansas, improvements in soil pH, phosphorus availability and aluminium stress indicate potential value in acidic Alfisol-based soybean and corn systems. In southwestern Kansas, the Entisol response is especially relevant because irrigated production depends heavily on groundwater from the High Plains aquifer, where long-term water-level decline continues to constrain agricultural sustainability (Kansas Geological Survey, 2024–2025; U.S. Global Change Research Program & NOAA Climate.gov, 2024–2025).

The increase in soil water retention and corn productivity under deficit irrigation therefore suggests potential value for water-limited systems. Nevertheless, the study does not yet demonstrate landscape-scale reductions in groundwater withdrawal. Such hydrological benefits would require evaluation at commercial farm and watershed scales.

The relevance of the Entisol findings is reinforced by the continuing decline in High Plains aquifer levels across western and southwestern Kansas.

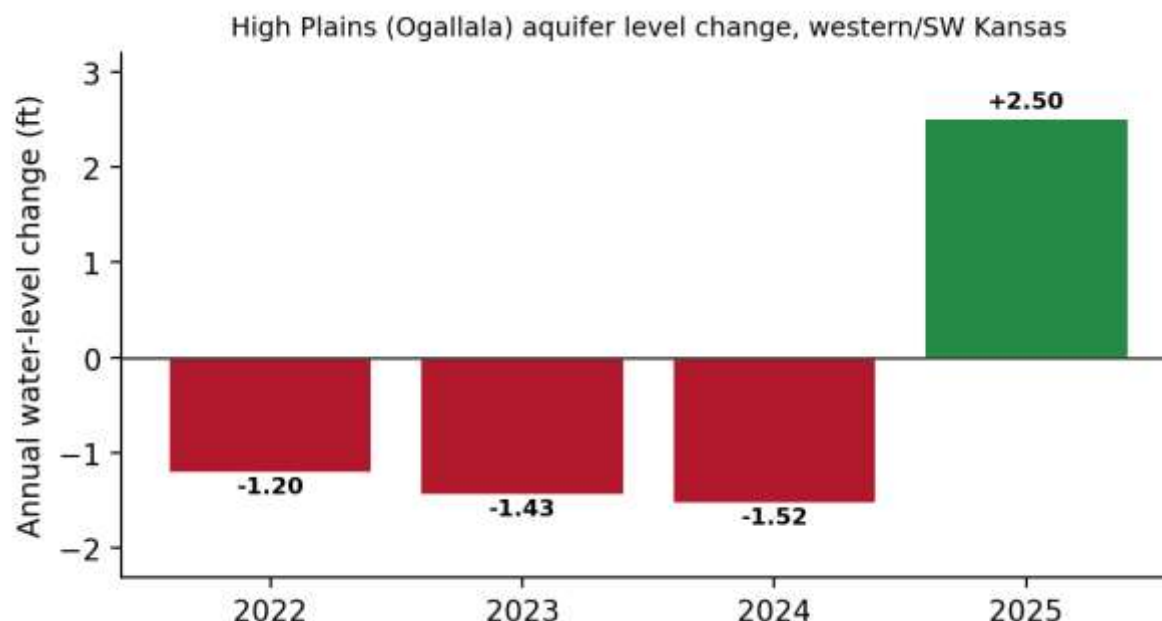


Figure 5: Recent annual water-level changes in the western and southwestern Kansas High Plains (Ogallala) aquifer

These conditions strengthen the practical importance of technologies that improve soil water retention and sustain crop productivity under reduced irrigation.

5.7 Limitations and Future Research

Several limitations should be considered when interpreting the findings. First, the principal field programme covered a single growing season and relied largely on replicated small-plot trials. Consequently, the reported changes in soil organic matter, microbial populations and soil structure require confirmation over multiple seasons and crop rotations.

Second, only three principal soil environments were evaluated, and responses may differ under other soil textures, climates, crop genotypes and management systems. Third, greenhouse stress experiments were conducted under controlled conditions and may not fully represent interacting field stresses. Fourth, the economic analysis was based on trial-level data and requires validation using commercial-scale costs and market conditions.

Future studies should therefore include multi-year, multi-location trials, commercial-scale validation, detailed greenhouse-gas measurements and life-cycle assessment. Additional research should also determine how formulation rate, microbial composition and hormone balance interact with soil type and crop requirements.

17. Conclusion

This 12-month integrated laboratory, greenhouse and multi-site field study establishes that a nano-encapsulated seven-strain microbial biostimulant consistently improves nutrient-use efficiency, reduces nutrient losses, rebuilds soil biological function and strengthens plant stress tolerance across three contrasting soil orders, with outcomes that reproduce and, in several indices, exceed an international benchmark. The chitosan carrier preserved a living consortium above 1×10^9 CFU mL^{-1} through temperate climatic extremes, and the resulting field responses—14–31% yield gains, 28% improved nitrogen-use efficiency, 25% fertilizer substitution without yield penalty, ~50% leaching reduction in sandy soils, and substantial single-season gains in soil organic matter and microbial biomass—define clear, evidence-based deployment pathways. The convergence of laboratory mechanism, greenhouse physiology and field performance provides a strong scientific foundation for scaling the technology as a soil-health and climate-resilience tool in Kansas and analogous Great Plains agricultural systems.

Declarations

Data availability: The datasets generated and analysed during the study, and the detailed one-year field-trial protocol (application rates, soil sampling depths, irrigation management and data-recording formats), are available from the corresponding author on reasonable request.

Author contributions E.M.R. designed the study, supervised the laboratory, greenhouse and field programme, performed data interpretation and wrote the manuscript. J.T.K. conducted field and laboratory trials, performed statistical analysis and contributed to manuscript preparation. Both authors read and approved the final manuscript.

Competing interests: The authors are affiliated with Aiden Digital Labs USA LLC, which develops the biostimulant technology described. The formulation and methods are the subject of patent filings.

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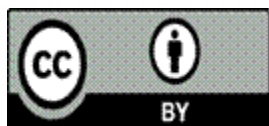
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